Propensity score matching: Dehejia and Wahba, and its replications

Can a nonexperimental study find the effect a randomised experiment found? Dehejia and Wahba (1999), Smith and Todd (2005) and Dehejia (2005), reproduced.

Tested against	Dehejia and Wahba (1999), *Journal of the American Statistical Association*, Tables 1 and 3; Smith and Todd (2005), *Journal of Econometrics*, Tables 3, 5 and 7; and Dehejia (2005), its reply, Tables 2–4. The data are LaLonde's NSW men and the PSID and CPS comparison groups from Dehejia's data page (read into memory and not saved). The matches were run again in R with `MatchIt` and `Matching`, and the balance checked with `cobalt`.
Missing in Julia	the logit propensity score, common support by the treated range or by Smith and Todd's density trimming, nearest-neighbour matching with replacement, local linear matching, the effect on the treated and the balance of the covariates. Julia has no counterpart of MatchIt, Matching or cobalt.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *Propensity score matching: Dehejia and Wahba, and its replications* (`PropensityMatching.ipynb`)
Result	123 of 145 printed values to every digit, the other 22 in the last. Smith and Todd's PSID logit to every digit.

The problem

LaLonde (1986) took the National Supported Work experiment, whose randomised controls give the effect of the training on the earnings of the men treated, about $1,794, and asked whether nonexperimental methods could find it from survey data instead: the PSID and CPS samples in place of the controls. They could not; the estimates ranged widely and depended on the specification. Dehejia and Wahba (1999) matched each treated man to the comparison men with the nearest estimated probability of treatment, the propensity score, and came close to the experiment. Smith and Todd (2005) replied that this works only in the subsample Dehejia and Wahba chose, and only for some specifications; Dehejia (2005) replied that the specification must be chosen for each sample by the balance of the covariates. The three papers together are a test of matching, and of how much an estimate depends on choices a paper does not print.

The method

The propensity score $p(x) = P(d = 1 \mid x)$ is fitted by a logit (`logit`), with earnings in thousands of dollars. Comparison units whose score lies outside the range of the treated are discarded (`commonsupport`), or, as Smith and Todd do, the treated whose score density among the comparison units is below the 2 per cent point are dropped (`trimmedsupport`, a quartic kernel with Silverman's bandwidth). Each treated unit is matched, with replacement, to the k comparison units with the nearest score (`nnmatch`), on p or on the log-odds $\ln(p/(1-p))$ (`logodds`), and the effect on the treated is the mean difference in 1978 earnings (`att`). When several comparison units are exactly as near, `nnmatch` takes the first in the data; `tierange` gives the smallest and largest effect over every way of breaking those ties. `locallinear` is Smith and Todd's local linear matching (their eq. 11), `matchweights` turns matches into weights and `balance` gives the standardised differences of the covariates, as `cobalt` does.

What the tests show

Test	What is compared	Agreement
DW Table 1	means of 8 covariates, 10 samples: 77 values	62 to every digit, 15 in the last
DW Table 3	col. (1), the unadjusted differences: 7 values	6 to every digit, 1 in the last
ST Table 3	DW 2002 logits, CPS: 15 coefficients, 15 SEs	29 to every digit, 1 in the last
ST Table 3	DW 2002 logit, PSID: 16 coefficients	every one
ST Tables 5, 7	bias of the mean difference and regression: 6 values	5 to every digit, 1 in the last
ST Table 5	CPS local linear matching, bandwidths 1 and 4	both in the last digit
Dehejia Tables 3, 4	nearest-neighbour effects: 7 values	5 to every digit, 2 in the last

The last-digit differences are mostly truncation: Dehejia and Wahba print 14016.8 as 14016 and 33.2252 as 33.22. The others, side by side (Julia against the paper):

- Smith and Todd's CPS intercept: −39.832651 vs −39.8326.
- Smith and Todd's CPS local linear bias: −87.0 vs −88 (bandwidth 1) and −67.94 vs −67 (bandwidth 4).

- Dehejia's ST specifications, nearest neighbour: 1,282.49 vs 1,283 (CPS) and 1,990.47 vs 1,991 (PSID).

Against R

With Julia's scores passed in, every match `MatchIt` makes is one of the exactly nearest comparison units, for all 185 or 297 treated in all 8 cases, and its effects equal Julia's where the tie choices coincide: 1,869.07, 1,990.47 and 862.72. `Matching` averages over near-ties within an internal tolerance that `distance.tolerance` does not switch off, so it gives 1,829.82 against 1,869.07 for PSID/DW; on CPS/Lalonde it gives 1,858.87, Julia's average over ties. `cobalt`'s `bal.tab` with Julia's weights gives the same 16 standardised differences to 4 decimals (age −1.2627 before matching and 0.0431 after, `re75` −5.4458 and −0.1997).

Three things the papers do not say

Ties decide the CPS estimate

In Dehejia's CPS/DW specification 22 treated men have two or more comparison men at exactly the same score, in sets of up to 9. Taking the first in the data gives 1,762; over every way of breaking the ties the effect runs from 1,055 to 1,964, and their average is 1,622. The printed 1,589 is inside that range, but no rule a reader could follow from the paper gives it, so it is not among the tests. Smith and Todd's CPS single-neighbour bias, 407, is likewise inside a tie range of 41 to 828 (Julia's first-of-ties gives 535).

A printed mean difference is the 1975 gap

Smith and Todd's Table 7 gives the PSID mean difference as −17,796 (846). That is the gap in 1975 earnings, the value of their own Table 8; the gap in 1978 earnings, the outcome, is −16,999, as in their Table 5.

A corrected mean

Dehejia and Wahba's Table 1 gives 0.18 for the share of Hispanics in PSID-3; the data give 0.117. The other 76 means agree, so the value is left untested; it looks like a correction to the data after the table was printed.

Still open

- **Dehejia and Wahba's own matches.** Their Table 3, columns (2)–(8), does not reproduce: PSID-1 gives 560 against 1,691. Smith and Todd (footnote 40) note that the sample on which the score is fitted moves these estimates by $200–300; the Dehejia–Wahba specifications are tested through Smith and Todd's logits and Dehejia (2005) instead.
- **Smith and Todd's other estimators.** Their PSID single-neighbour bias, 361, is what matching on *p* gives; on the log-odds they say they use, it is 319. The ten-neighbour and trimmed estimators come within about $100 (CPS ten neighbours −76 vs −5, inside a tie range of −176 to 14), and the PSID local linear bias does not reproduce (−160 and 124 vs −122 and 143). The tie rule and the trimming bandwidth are not stated.
- **Smith and Todd's PSID standard errors** agree only to about three significant figures (intercept 2.0623 vs 2.0629), and are not tested.
- **The Lalonde and ST samples.** Dehejia's cross-specifications on LaLonde's sample need 1974 earnings, which the public files do not have for it, and Smith and Todd's early random assignment sample cannot be rebuilt from them.

Further considerations

- **Report the tie range.** Where a score takes few values, as here, an estimate can depend on the order of the data; `tierange` shows by how much, and is worth reporting beside any nearest-neighbour estimate.
- **Other matching estimators.** Kernel matching, calipers and Abadie and Imbens's bias-corrected standard errors are in `Matching`; each would need its own published table before it becomes a test here.
- **Upstream.** Julia has no counterpart of MatchIt or cobalt; the three papers are a ready-made test for one.

References

Dehejia, R. H. and Wahba, S. (1999). Causal effects in nonexperimental studies: reevaluating the evaluation of training programs. *Journal of the American Statistical Association* 94(448), 1053–1062. doi:10.1080/01621459.1999.10473858

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Smith, J. A. and Todd, P. E. (2005). Does matching overcome LaLonde's critique of nonexperimental estimators? *Journal of Econometrics* 125(1–2), 305–353. doi:10.1016/j.jeconom.2004.04.011

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Data: Dehejia's NSW, PSID and CPS files, from Dehejia's data page at the NBER, read into memory and not saved, since they are not offered for redistribution with Pollis.

Methods written and tested with Claude Code (Anthropic) in Pollis, from the published papers and their data.

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